Does a quantum kernel beat a classical one on real data, when the comparison is fair?

Quantum-kernel SVMs are the standard first QML example: encode each sample with Qiskit's ZZ feature map (Havlíček et al. 2019), use state overlaps as the kernel, train an SVM. Tutorials often show good accuracy on small tabular datasets, but rarely against a **tuned** classical model on the **same features and splits**.

Why it matters

- A newcomer can't tell whether a result comes from the quantum kernel or from the preprocessing, the split, or an untuned baseline.
- Two known effects, bandwidth (Shaydulin & Wild 2022) and exponential concentration (Thanasilp et al. 2022), decide whether the kernel learns at all. We measure both on the same data.
- Hardware cost is rarely shown: one circuit per *pair* of samples, under realistic noise.

Data (small, real, public)

Dataset	Samples	Features	Positive class
breast cancer	569	30	62.7%
wine	178	13	33.1%

Breast cancer: Wisconsin diagnostic, malignant vs benign. Wine: cultivar 1 vs the other two. Both ship with scikit-learn.

Short answer

No. Tuned, the quantum kernel ties the RBF SVM within noise but is never ahead (0 of 12 settings). Untuned, it collapses as qubits grow. The best model overall is classical.

One protocol for every model: same features, same splits, same tuning budget

1 · Outer split
3x repeated stratified 5-fold: 15 train/test folds, identical for every model.

→

2 · Features
StandardScaler → PCA(n = qubits) → MinMax to [0, π], fitted on the training fold only.

→

3 · Inner CV
Stratified 3-fold on the training fold picks C, RBF γ and quantum bandwidth s.

→

4 · Test once
Refit on the training fold, score on the untouched test fold.

→

5 · Paired test
Per-fold differences, Nadeau & Bengio (2003) corrected resampled t-test.

Five models, same inputs

QK tuned
s ∈ {0.01, 0.02, 0.05, 0.1, 0.2, 0.4, 1}

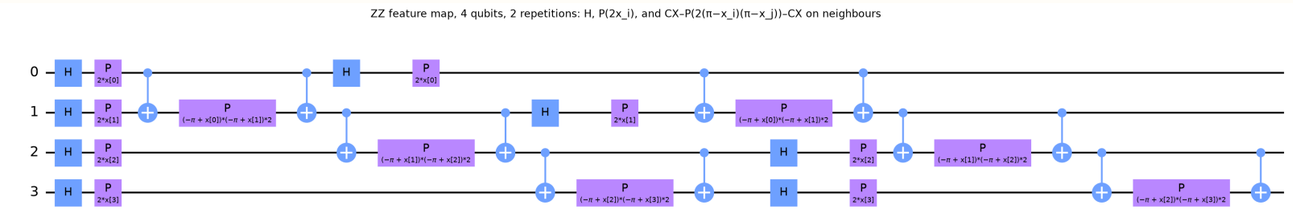
QK s = 1
tutorial scaling

RBF SVM
γ ∈ 6 values

LogReg
C tuned

RBF SVM
all original features

Quantum kernel $k(x, y) = |\langle\phi(sx)|\phi(sy)\rangle|^2$, ZZ feature map, 2 reps, linear entanglement. $C \in \{0.1, 1, 10, 100\}$ for every SVM. Qubits n = 2, 4, 6, 8, 10, 12.

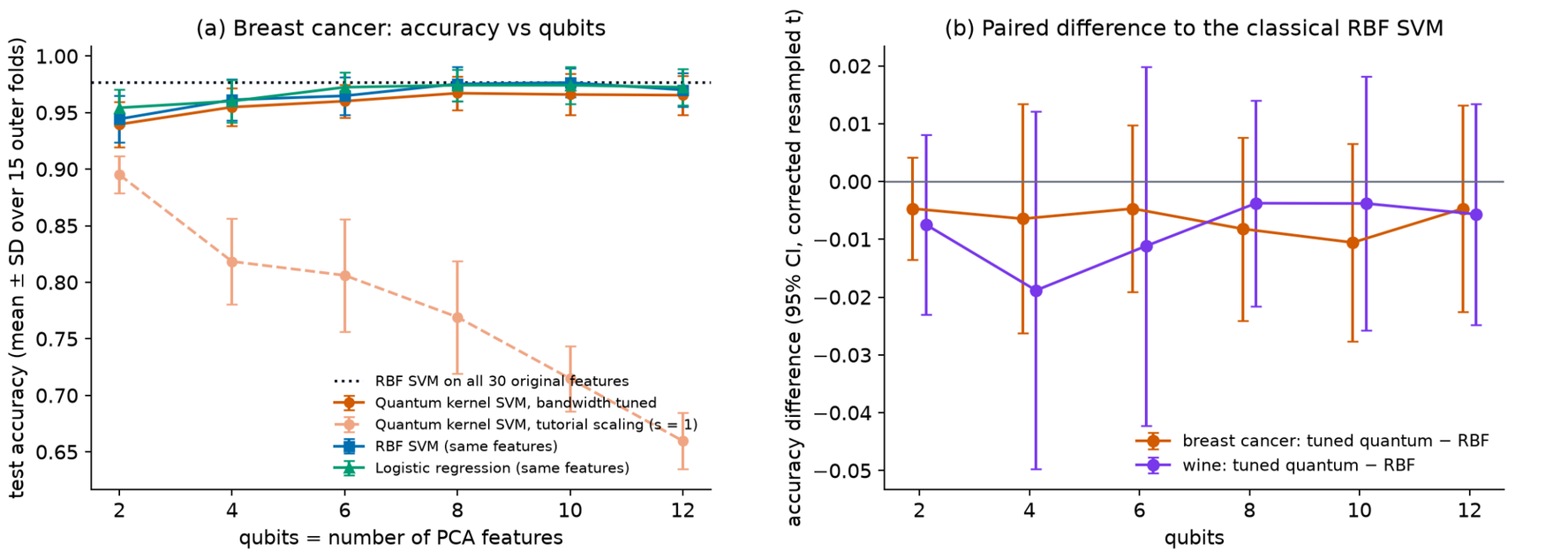


Engineering

- Batched NumPy statevectors: a 12-qubit kernel in seconds. Tests check they equal Qiskit's `zz_feature_map` statevectors.
- Compute-uncompute overlap circuits for shots and noise; tests check P(0…0) equals the kernel.
- A test proves preprocessing is fitted on the training fold only (no leakage).
- Concentration study and geometric difference g (Huang et al. 2021) are unsupervised: labels unused.

Classical simulation only — no quantum hardware · NumPy statevector (checked against Qiskit), Qiskit Aer, local FakeTorino noise model

The tuned quantum kernel never beats the classical RBF SVM



0 / 12

settings where tuned QK beats RBF SVM. Mean differences −1.9 pts to −0.4 pts; 12 of 12 CIs include 0 (smallest p = 0.207).

89.5% → 66.0%

QK at tutorial s = 1, breast cancer, 2 → 12 qubits (wine 88.3% → 66.9%). Worse than RBF in 12 of 12 settings at $p < 0.05$.

97.7%

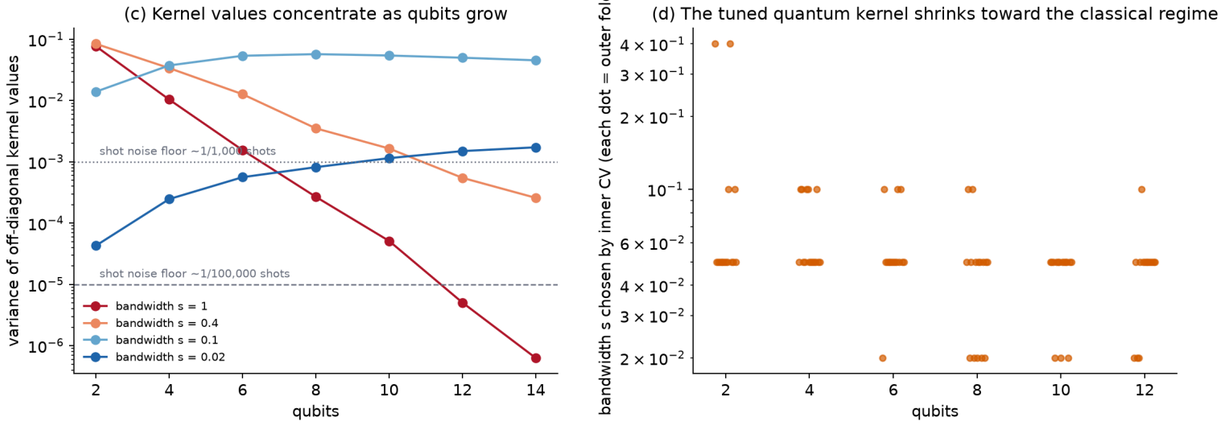
Best breast-cancer models: RBF SVM on 10 PCA features and RBF SVM on all 30 original features (tied). Best quantum: 96.7% at 8 qubits.

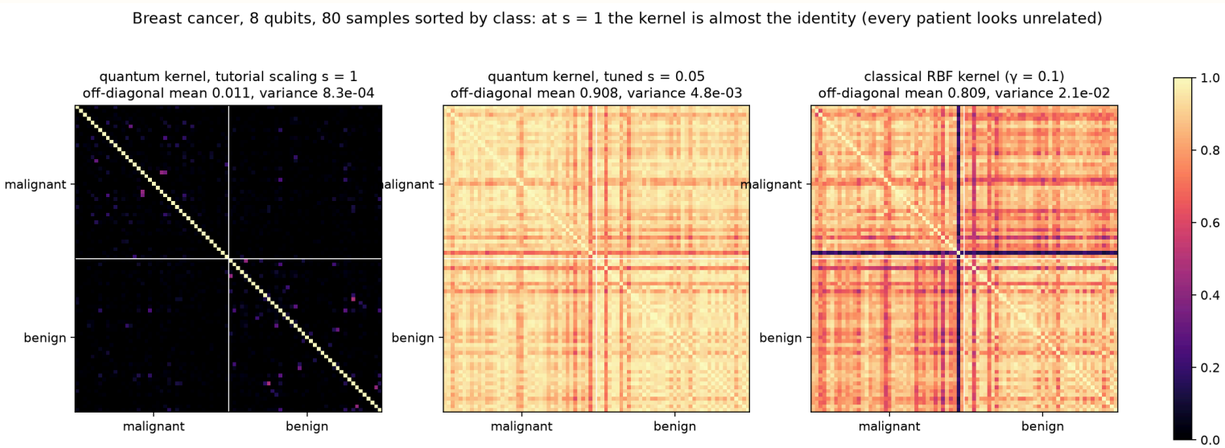
99.8%

Best wine model: RBF SVM on 12 PCA features. Best quantum: 99.2% at 12 qubits.

The quantum kernel's only nominally significant win anywhere: vs logistic regression on wine, 2 qubits (+3.2 pts, p = 0.045; 1 of 12 uncorrected tests, against a linear baseline).

Concentration at s = 1, and a tuned bandwidth that makes the kernel classical

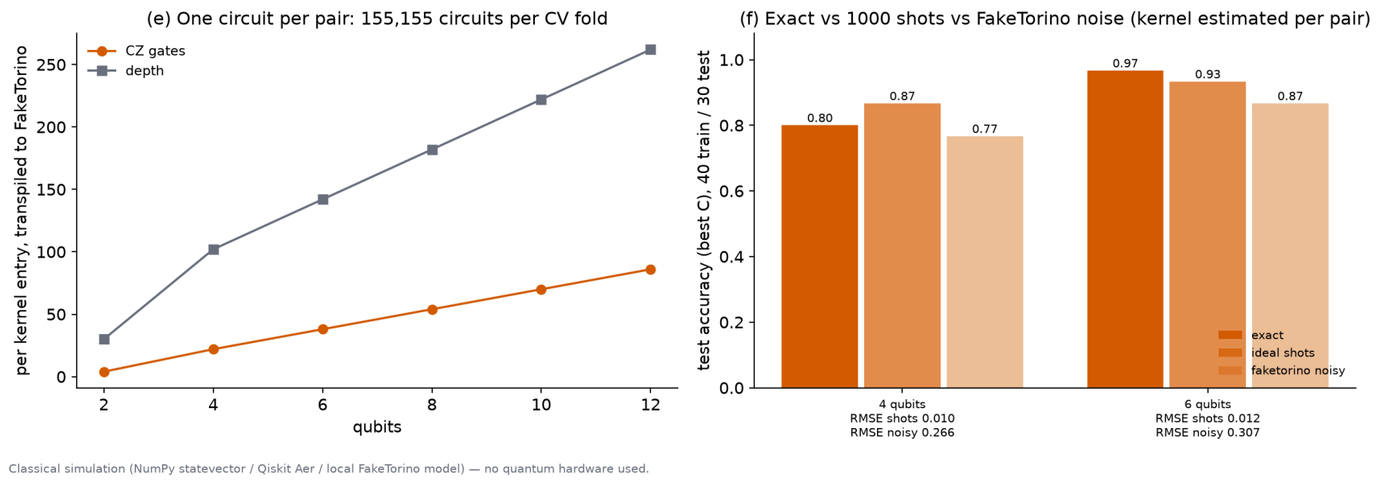




Classical simulation (NumPy statevector, checked equal to Qiskit) — no quantum hardware used.

- **Concentration.** At s = 1 the variance of off-diagonal kernel values falls from 7.7×10^{-2} (2 qubits) to 6.3×10^{-7} (14), about 2.7× per qubit (Thanasilp et al. 2022). Below a ~1/1,000-shot floor from 8 qubits, ~1/100,000 from 12: the kernel becomes identity plus shot noise.
- **Bandwidth.** Inner CV picks a small scaling: median s = 0.05; s ≤ 0.1 in 158 of 180 outer-fold choices, s = 1 in 0. The fix Shaydulin & Wild (2022) describe, and it works.
- **But small s is classically easy.** At s ∈ {0.02, 0.1}, bracketing the CV choice, the geometric difference to the closest RBF kernel is only g = 1.2–3.1 (Huang et al. 2021): an RBF kernel already captures it. The largest g (28, s = 1, 2 qubits) belongs to a model that scores 89.5% vs 94.4% for RBF. Large g is necessary, not sufficient.

On a device, every kernel entry is a noisy circuit, and there are a lot of them



Subset (1,980 circuits, 1,000 shots)	RMSE shots	RMSE noisy	mean k exact → noisy	accuracy exact / shots / noisy
4 qubits, s = 0.05	0.010	**0.266**	0.86 → 0.60	80 / 87 / 77%
6 qubits, s = 0.05	0.012	**0.307**	0.79 → 0.48	97 / 93 / 87%

40 train / 30 test breast-cancer samples, one split; accuracy uses the best C on the test set (optimistic; 1 sample = 3.3 pts).

- **Shots are not the problem.** 1,000 shots give kernel RMSE 0.010–0.012.
- **Noise is.** The FakeTorino noise model (historical calibration snapshot, run locally in Aer) gives RMSE 0.27–0.31, about 25× the shot error. Noise pulls P(0…0) down, so every pair looks less similar.
- **Circuit count.** One overlap circuit per pair: 103,285 train + 51,870 test = **155,155** per CV fold on the full dataset. With 7 bandwidths and 15 outer folds, our protocol at one qubit count needs ≈ 11.6 M circuits (12 billion shots).
- **Circuit depth.** Transpiled to FakeTorino, one entry costs 4–86 CZ (depth 30–262) for 2–12 qubits.

Takeaways, honest limits, and how to reproduce

Takeaways

- Under a fair protocol the quantum kernel **ties at best**: 0 of 12 settings ahead of the RBF SVM; the best model is classical on both datasets.
- Untuned (s = 1), kernel values concentrate and accuracy falls with every qubit. **Always tune the bandwidth**.
- Tuning works by making the kernel nearly classical (g ≈ 1–3). Check g before claiming an edge.
- On a noisy device the kernel is distorted ~25× more by noise than by shots, at 155k circuits per fold.

Honest limits

- Two small tabular datasets, binary tasks. One feature map (ZZ, 2 reps, linear); no trainable or projected kernels.
- PCA to n qubits discards information; finite hyperparameter grids.
- Noise study: one 40/30 split, 2 qubit counts, one historical noise model, no error mitigation.
- 24 paired tests, not corrected for multiplicity.
- Everything is classical simulation. No IBM account, no hardware. **No quantum advantage is claimed.**

Reproduce

```
uv venv -p 3.12 .venv
uv pip install -p .venv/bin/python \
  -r requirements.lock.txt
uv pip install -p .venv/bin/python \
  -e . --no-deps
# ~3 min on a laptop:
python -m qkbench.experiments
python -m qkbench.figures
python -m qkbench.visualize
python presentation/build.py
python -m pytest -q
python -m streamlit run app/app.py
```

Python 3.12.13 · Qiskit 2.6.0 · Aer 0.17.2 · scikit-learn 1.9.1 · NumPy 2.5.4. Artifacts 2026-10-25T12:08:24Z. Every number here is generated from `artifacts/*.json`. MIT licence.